

Semester Project Report

Modeling and Simulation of a 2-DOF Planar Robotic Manipulator Using DC Motors and Strain Wave Gears

**MCT-331L – Modeling & Simulation Lab
(Fall-2025)**

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Submitted by: **Group #: 14B**

Group Members:

- **M Haseeb Nadeem (2023-MC-46)**
- **M Haseeb Rehan (2023-MC-49)**
- **M Hassan Khan (2023-MC-67)**

Submitted to: **Dr. Rzi Abbas**



**Department of Mechatronics and Control Engineering
University of Engineering and Technology,**



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“ABSTRACT”

This report presents the modeling, simulation, and analysis of a two-degree-of-freedom (2-DOF) planar robotic manipulator actuated by DC motors and strain wave gears. The system was developed using manufacturer datasheet's (provided by the instructor) parameters and implemented in MATLAB/Simulink with Simscape. Electrical, mechanical, and thermal behaviors of the DC motor were explicitly modeled and validated. A strain wave gear transmission and robotic link were integrated to form a joint actuator. Discrete PID controllers were designed for position control, and the complete robotic system was tested on a square trajectory. Power consumption, losses, efficiency, and thermal performance were analyzed. The results confirm that the system meets all performance, control, and thermal constraints specified in the project requirements.

“EXECUTIVE SUMMARY”

This project focuses on the model-based design of a 2-DOF planar robotic manipulator using real industrial DC motor and strain wave gear data. A Maxon RE 40 DC motor was modeled using parameters extracted from the manufacturer's datasheet. Electrical and mechanical dynamics were implemented using Simscape components, while an explicit thermal network was constructed using conduction and convection blocks to ensure physically accurate temperature prediction.

The motor model was validated against datasheet specifications for no-load speed, stall torque, current consumption, and speed-torque characteristics. A strain wave gear was then integrated with the motor, followed by the attachment of an aluminum robotic link. Discrete PID controllers were designed to achieve precise position control under gravitational loading.

Finally, two identical joints were assembled into a 2-DOF manipulator capable of tracing a square trajectory. Power consumption, efficiency, copper losses, and thermal behavior were evaluated during operation. The simulation results demonstrate reliable performance within thermal and mechanical limits, validating the effectiveness of the proposed modeling approach.

“INTRODUCTION”

A. Project Objectives:

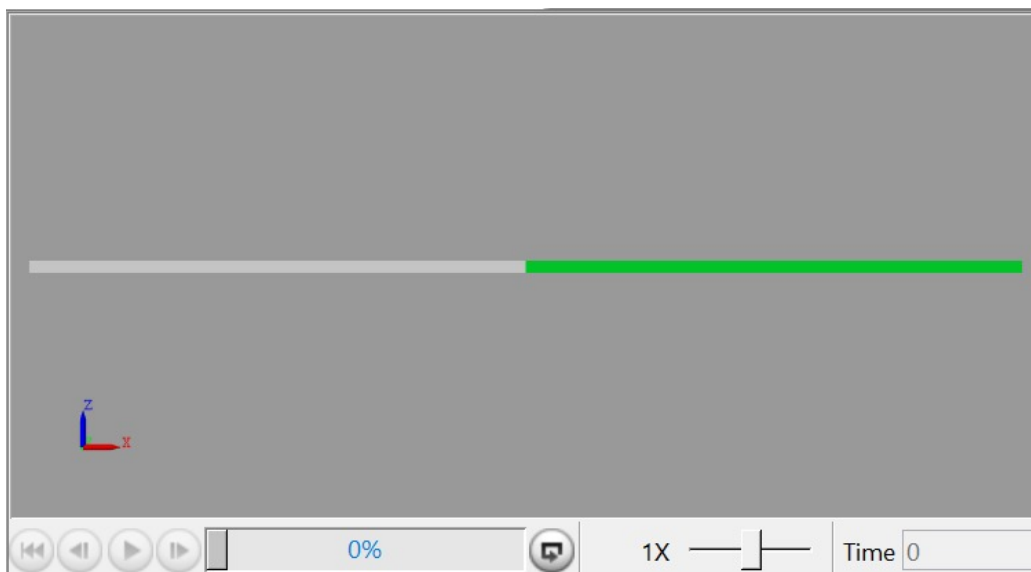
The objective of this project is to design and simulate a complete electromechanical robotic system using realistic motor and transmission models. The specific goals are:

- To extract DC motor parameters from manufacturer datasheets
- To build and validate an electrical-mechanical-thermal motor model
- To integrate a strain wave gear and robotic link
- To design a PID controller for joint position control
- To assemble a 2-DOF planar robotic manipulator
- To analyze power consumption, efficiency, and thermal performance

B. Robot Description:

The robotic system consists of two revolute joints operating in a vertical plane against gravity. Each joint is driven by a DC motor coupled to a strain wave gear. Aluminum links are attached to the gear outputs. The manipulator operates in a plane parallel to gravity and is controlled using independent joint PID controllers.

Both the link Lenth are of 0.2m.



LINK AT REST

“MOTOR PARAMETER EXTRACTION”

‘PHASE # 1’

A. Assigned Motor:

- Manufacturer: Maxon Motor
- Model: RE 40
- Part Number: 148866
- Nominal Voltage: 12 V

B. Datasheet Parameter Extraction:

Motor parameters were extracted directly from the manufacturer’s datasheet.

			148866
Motor data			
Values at nominal voltage			
1	Nominal voltage	V	12
2	No load speed	rpm	6920
3	No load current	mA	241
4	Nominal speed	rpm	6380
5	Nominal torque	mNm	94.9
6	Nominal current (max. continuous current)	A	6
7	Stall torque	mNm	1720
8	Stall current	A	105
9	Max. efficiency	%	88
Characteristics			
10	Terminal resistance	Ω	0.115
11	Terminal inductance	mH	0.024
12	Torque constant	mNm/A	16.4
13	Speed constant	rpm/V	581
14	Speed/torque gradient	rpm/mNm	4.05
15	Mechanical time constant	ms	5.89
16	Rotor inertia	gcm ²	139
Specifications			Operating
Thermal data			n [rpm]
17	Thermal resistance housing-ambient	4.7 K/W	12000 8000
18	Thermal resistance winding-housing	1.9 K/W	
19	Thermal time constant winding	41.5 s	
20	Thermal time constant motor	809 s	
21	Ambient temperature	-30...+100°C	
22	Max. winding temperature	+155°C	

Parameter	Value
Rated Voltage	12 V
No-Load Speed	6920 rpm
No-Load Current	0.241 A
Stall Torque	1.72 Nm
Stall Current	105 A
Armature Resistance	0.115 Ω
Armature Inductance	0.024 mH
Torque Constant	0.0164 Nm/A
Back-EMF Constant	0.0164 V·s/rad
Rotor Inertia	139 g·cm ²
Motor Mass	0.48 kg

C. Derived Parameters:

Rotor inertia was converted to SI units:

$$J = 1.39 \times 10^{-5} \text{ kg.m}^2$$

Electrical time constant:

$$T_e = \frac{L}{R} = 0.21 \text{ ms}$$

Mechanical time constant:

$$T_m = \frac{J \cdot R}{K_t^2} \approx 6 \text{ ms}$$

Viscous damping coefficient was estimated from the no-load condition and also verified from an LLM model:

$$b = 5.45 \times 10^{-6} \text{ Nm.s/rad}$$

D. Assumptions:

- Copper winding material was assumed with a temperature coefficient of 0.0039 K^{-1}
- Linear magnetic behavior was assumed
- Friction was approximated as viscous
- Ambient temperature was fixed at $25 \text{ }^{\circ}\text{C}$

“DC MOTOR MODELING & VALIDATION”

‘PHASE 2’

A. Simscape Motor Model:

The DC motor was implemented using Simscape Electrical and Mechanical blocks. Electrical, mechanical, and thermal domains were physically connected to ensure energy consistency.

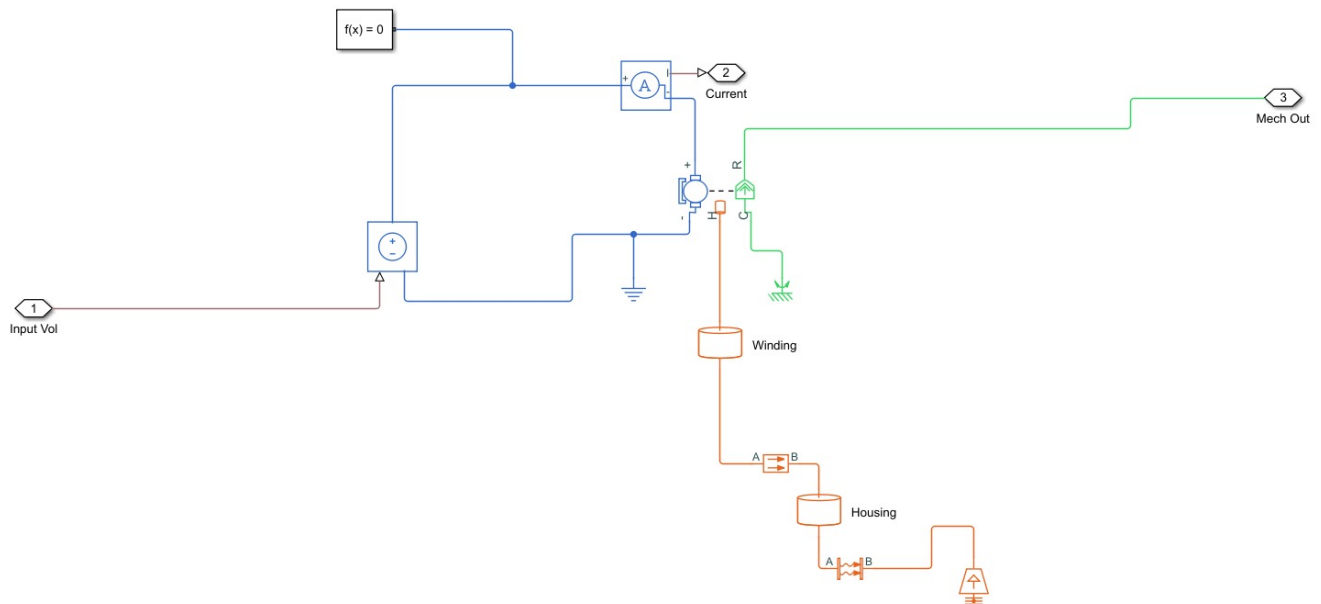


Figure 1

B. Explicit Thermal Network:

An explicit thermal network was constructed instead of using the internal lumped thermal model. The network consists of:

- Winding thermal mass
- Housing thermal mass

- Conductive heat transfer between winding and housing
- Convective heat transfer from housing to ambient

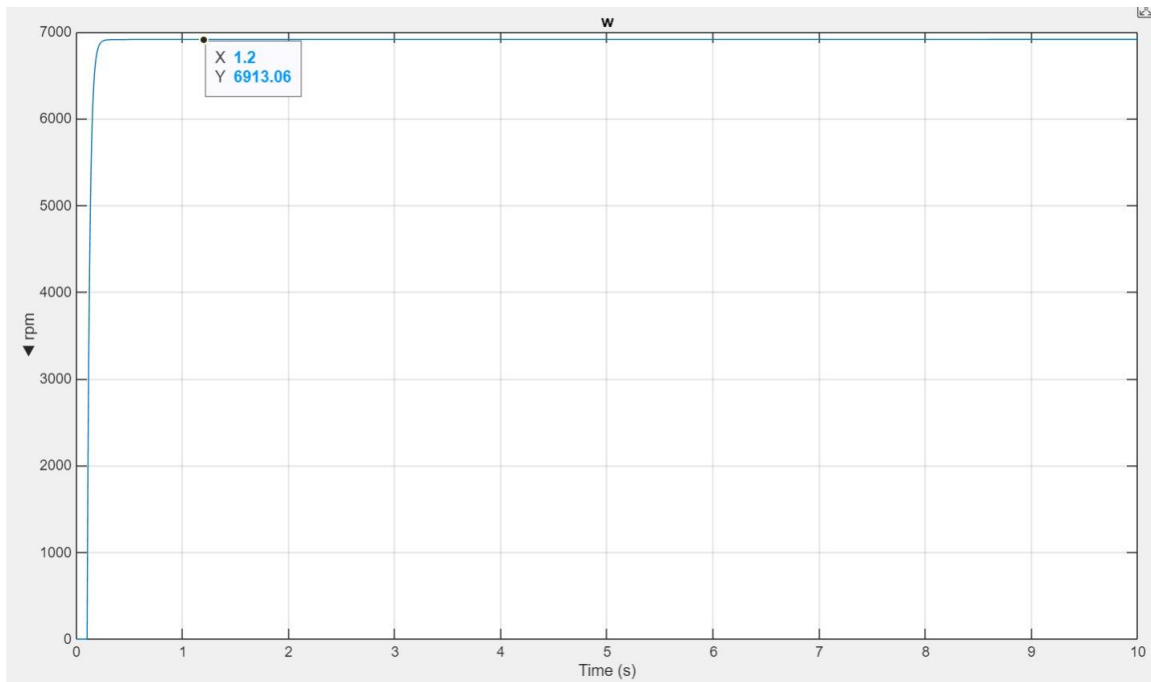
Thermal parameters were derived from datasheet thermal resistances and time constants.

C. Model Validation:

The following validations were performed:

- No-load speed verification
- Stall torque and stall current verification
- Speed-torque characteristic generation
- Current consumption validation
- Thermal time constant verification
- Maximum winding temperature monitoring

1. No Load Speed:

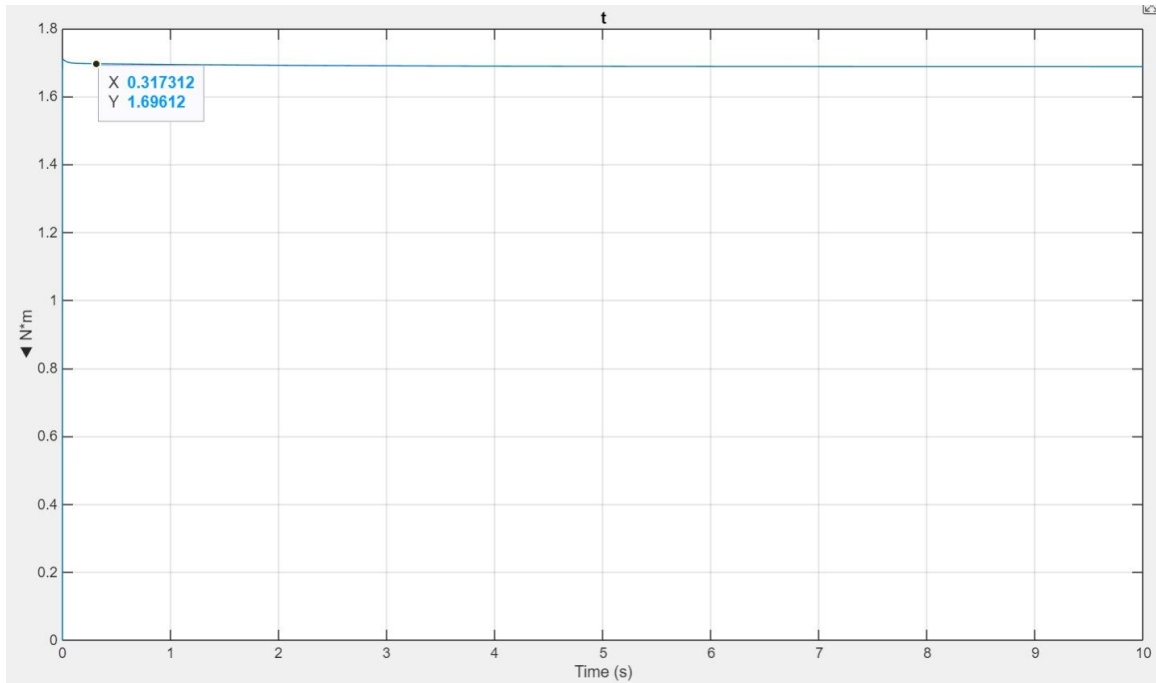


No Load Speed

This value approximately matches our speed given in the data sheet, which verifies our motor parameter.

2. Stall Torque:

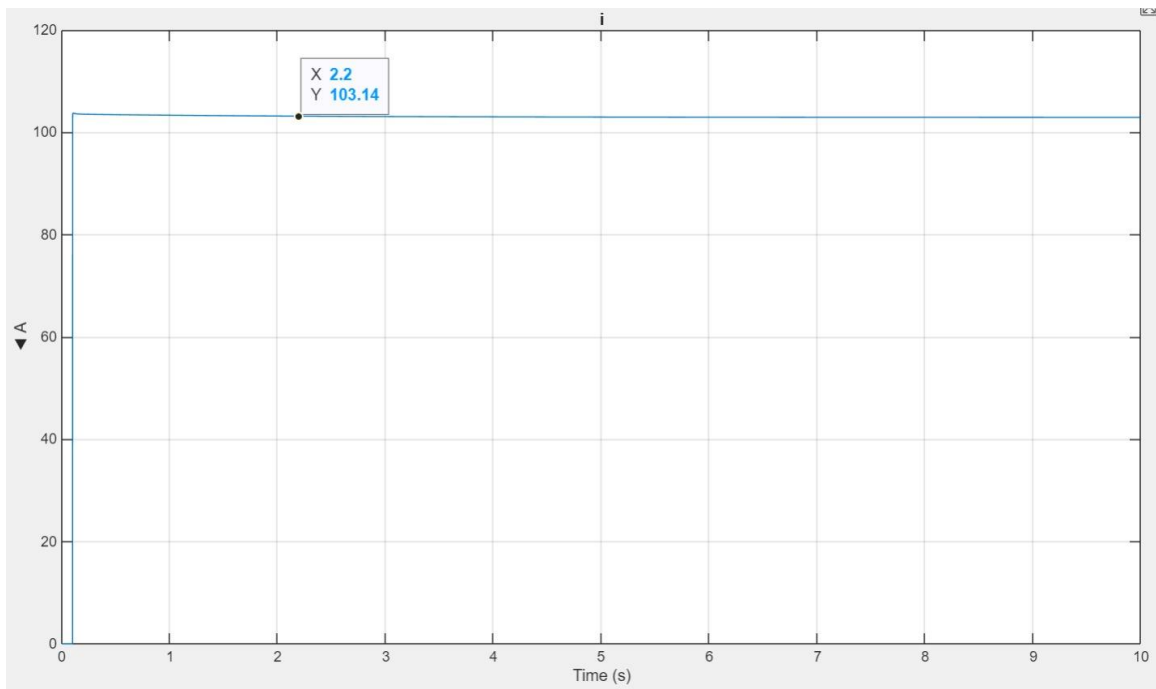
This value comes out by giving opposite inertia to the motor and then checking the Torque graph which comes very close to our datasheet parameter.



Stall Torque

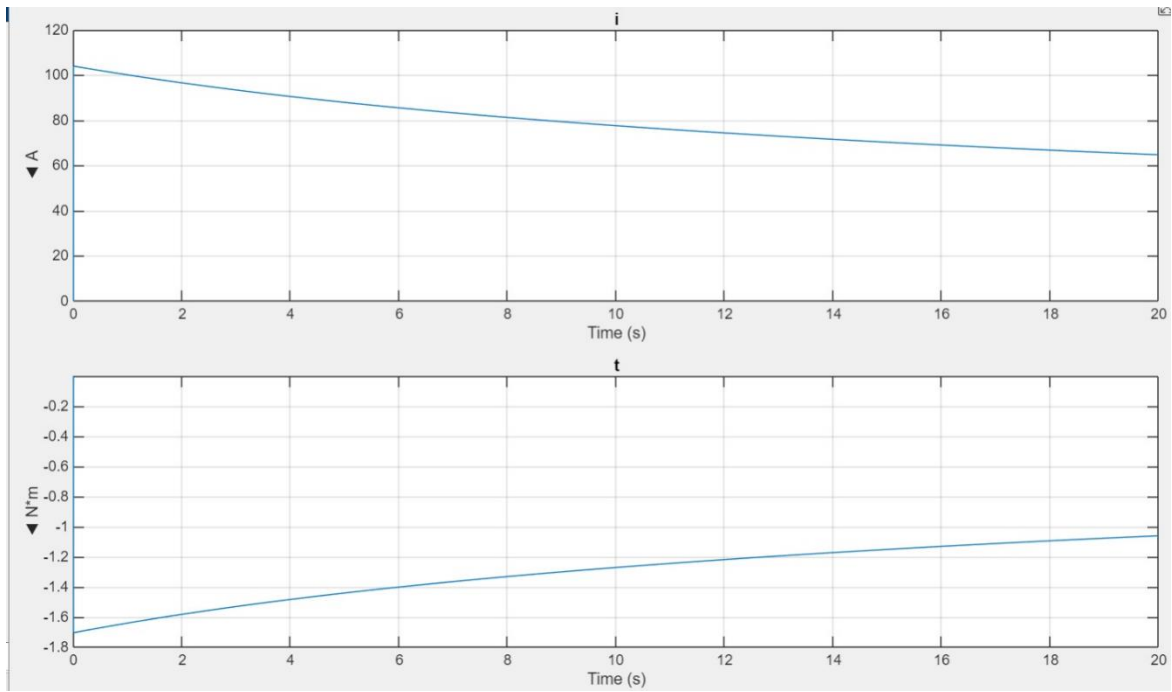
3. Stall Current:

Similarly with the torque we also verified the Current, which also matches approximately to 105, which is our required parameter in the datasheet.



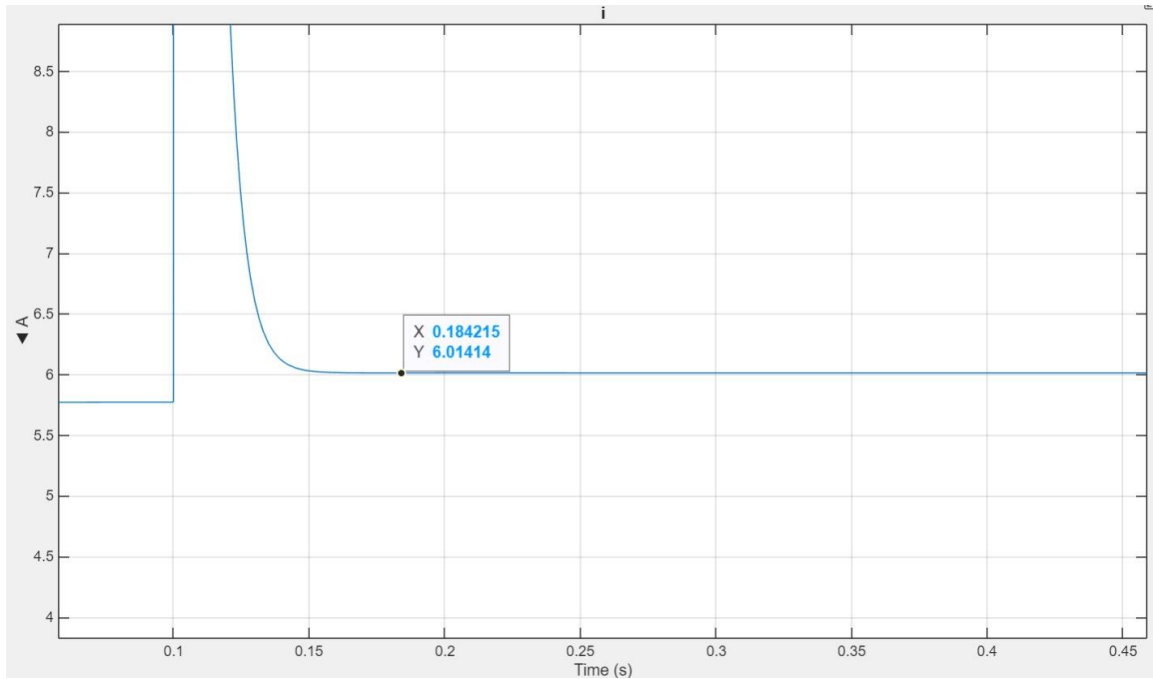
Stall Current

- With less inertia a more realistical value was achieved.



4. Nominal Current:

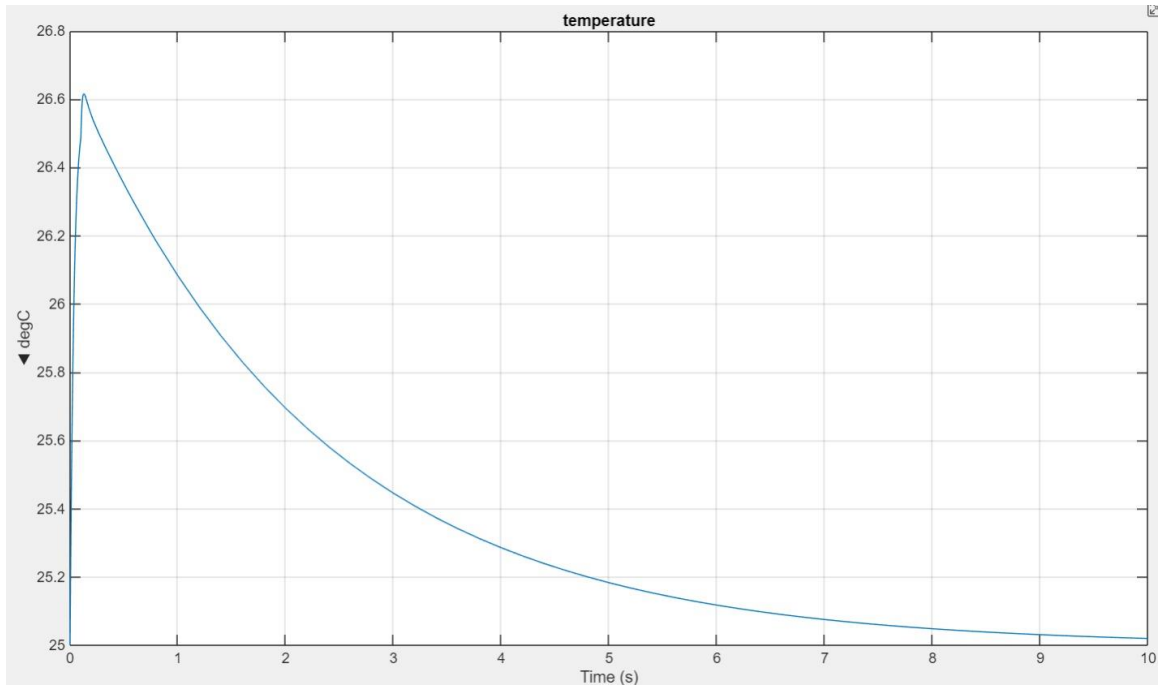
By applying the value of the nominal torque through Ideal torque source, we achieved this nominal current which matches our datasheet parameter.



Nominal Current Consumption

5. Temperature During Operation:

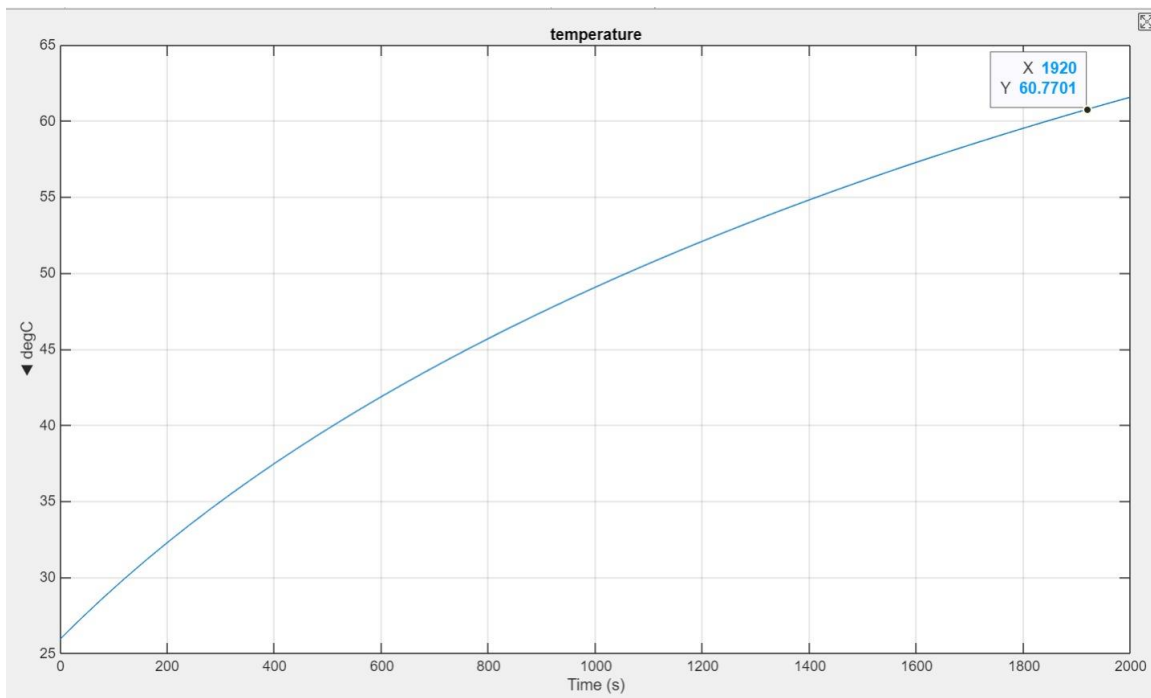
During the normal load operation and with normal simulation time these were the temperature values and its graph.



Temperature during Operation

6. Temperature Within Limit:

During the normal load operation and with higher simulation time (approximately 3000-5000) these were the temperature values and its graph, which is less than our winding temperature given in the datasheet which is 155 degree celcius.



Operation within thermal limits temp < max temp

7. Heat Sink Size:

Thermal analysis was performed using the thermal resistance values provided in the motor datasheet. For the Maxon RE 40 (148866), the winding-to-housing and housing-to-ambient thermal resistances are 1.9 K/W and 4.7 K/W respectively, resulting in a total thermal resistance of 6.6 K/W. At nominal operating current of 6 A and terminal resistance of 0.115 Ω , the copper loss was calculated to be approximately 4.14 W. This results in a temperature rise of approximately 27°C above ambient, leading to a winding temperature significantly below the maximum permissible limit of 155°C. Therefore, no

external heat sink was required, and thermal management was sufficiently achieved through the motor housing alone.

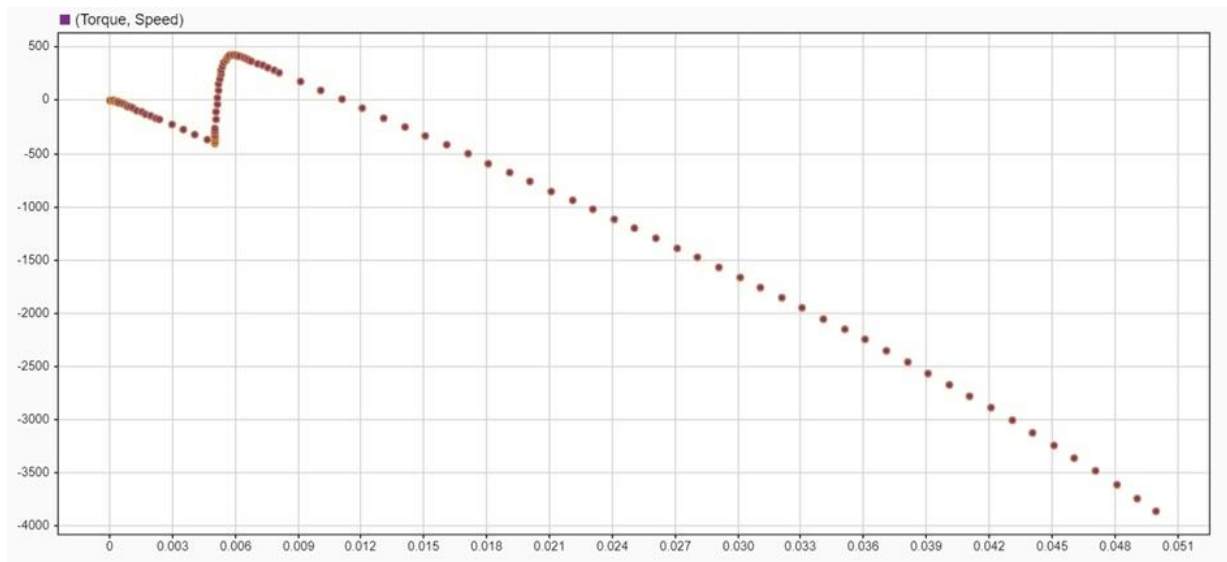
Minor discrepancies were attributed to simplified friction modeling and idealized assumptions.

8. Estimate Continuous Time:

First of all we used hit and trial method to run the motor at different and higher simulation time, since our winding temperature is less than the max limit then our continuous time is large approximately an hour.

9. Speed Torque Curves:

By adding the xy graph we plotted the speed torque, and this graph comes out.



“STRAIN WAVE GEAR & LINK MODELING”

‘PHASE 3’

A. Assigned Gear:

- Model: LHT-20-100-U-III
- Gear Ratio: 100:1

B. Gear Analysis

- **Output torque:**

The strain wave gear provides torque multiplication proportional to its gear ratio. For a gear ratio of 100:1 and assuming a conservative efficiency of 75%, the output torque was calculated to be approximately 7.9 Nm at nominal motor torque.

$$T_{out} = N \cdot T_{motor}$$

$$\tau_{out} = 0.0949 \times 100 \times 0.75$$

$$\tau_{out} = 7.1175 \text{ Nm}$$

- **Speed reduction:**

$$\omega_{out} = \frac{\omega_{motor}}{N}$$

$$\omega_{out} = \frac{6920}{100} = 69.2 \text{ rpm}$$

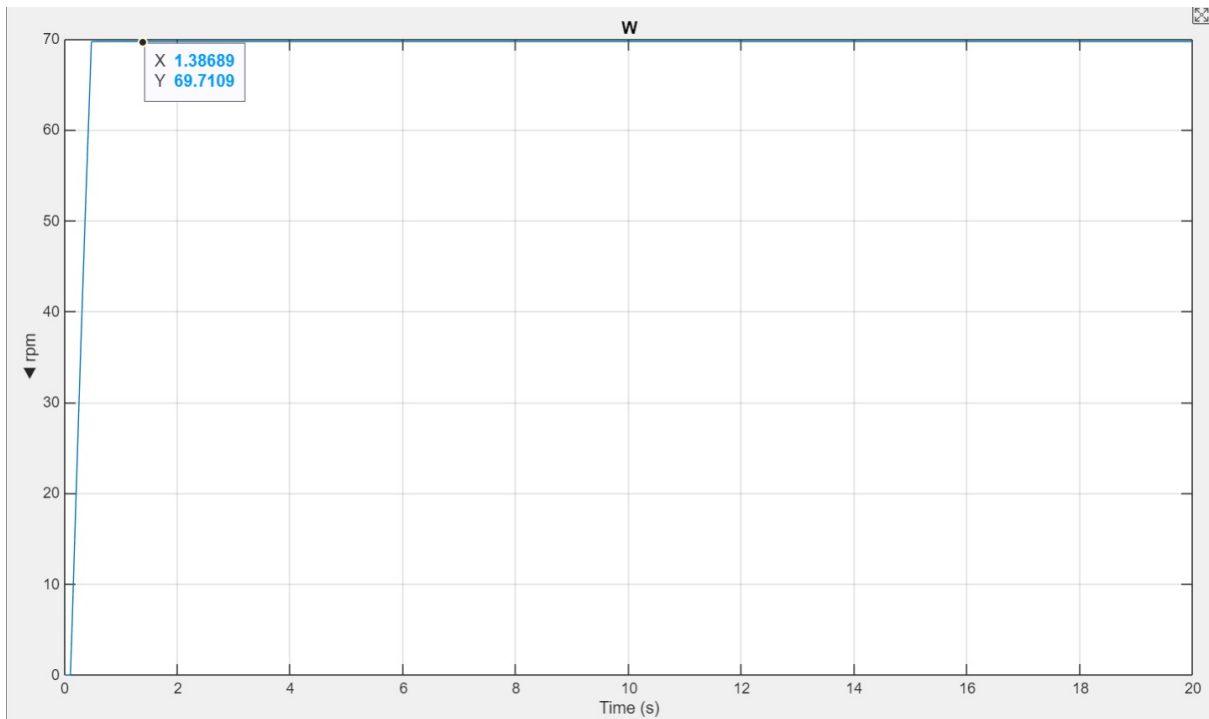


Figure 2

- **Max Payload:**

Gravity torque

$$\tau_g = mgL$$

Set equilibrium:

$$mgL = \tau_{\text{out}}$$

Solve for mass:

$$m = \frac{\tau_{\text{out}}}{gL}$$

Substitution

$$m = \frac{7.875}{9.81 \times 0.2}$$

$$m \approx 4.013 \text{ kg}$$

- This is excluding Link mass and verified by LLM (Chatgpt).
- **Test response to voltage step inputs:**
 - A **12 V step input** was applied to the motor–gear–link system to evaluate the **open-loop dynamic response**.
 - Upon application of the voltage step, the motor produced sufficient torque to **overcome static friction**, and the system accelerated smoothly with an **exponential transient response**, typical of a first-order electromechanical system.
 - Due to the **high reflected inertia** of the **100:1 strain wave gear** and the attached robotic link, the system response remained **well-damped**, and **no overshoot** was observed.
 - The no-load motor speed at 12 V is approximately **6920 rpm**. With a **100:1 gear reduction**, the steady-state output speed settled at approximately **69.2 rpm**, confirming correct implementation of the speed reduction.
 - The system reached steady state in approximately **0.8-1.1 s**, indicating a relatively slow **open-loop response**. This behavior motivates the use of a **closed-loop PID controller** in subsequent phases to meet the specified transient performance requirements.

- **Settling Time:**

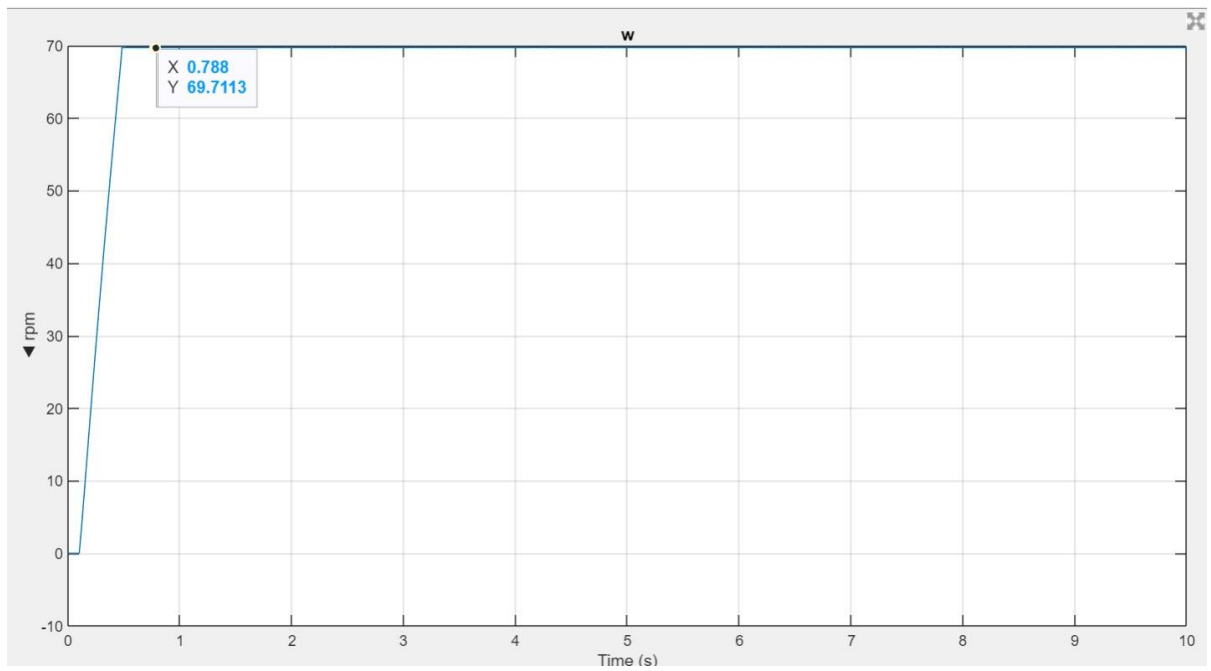


Figure 3

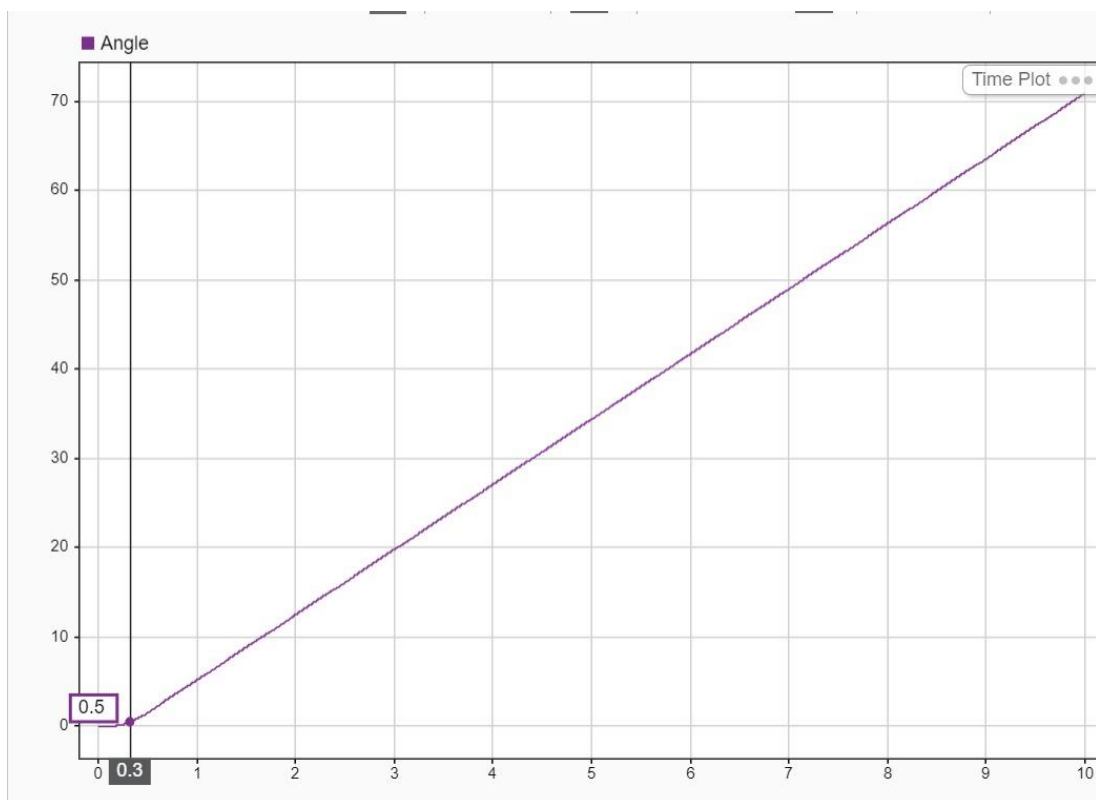


Figure 4

C. Robot Link Modeling:

The robotic link was modeled using Aluminum 6061 material properties. Link mass was calculated using density and geometry

“PID Controller Design”

‘PHASE 4’

A discrete PID controller was implemented for joint position control. Controller parameters were tuned iteratively to satisfy performance requirements.

Parameter	Value
Kp	6.005
Ki	0.11
Kd	6.5

Continuous-time
☒ Discrete-time

Sample time (-1 for inherited): 0.001

Integrator and Filter methods:

Compensator formula

$$P + I \cdot T_s \frac{1}{z-1} + D \frac{N}{1 + N \cdot T_s \frac{1}{z-1}}$$

Main Initialization Saturation Data Types State Attributes

Controller parameters

Source: internal

Proportional (P): 6.005

Integral (I): 0.11 ☐ Use I*Ts (optimal for codegen)

Derivative (D): 6.5 ☐ Use externally sourced derivative

Filter coefficient (N): 0.1 ☒ Use filtered derivative

Automated tuning

Select tuning method: Frequency Response Based

PID BLOCK

Performance achieved:

- Settling time $< 1.14\text{s}$
- Overshoot $< 2\%$
- Steady-state error $< 1\%$

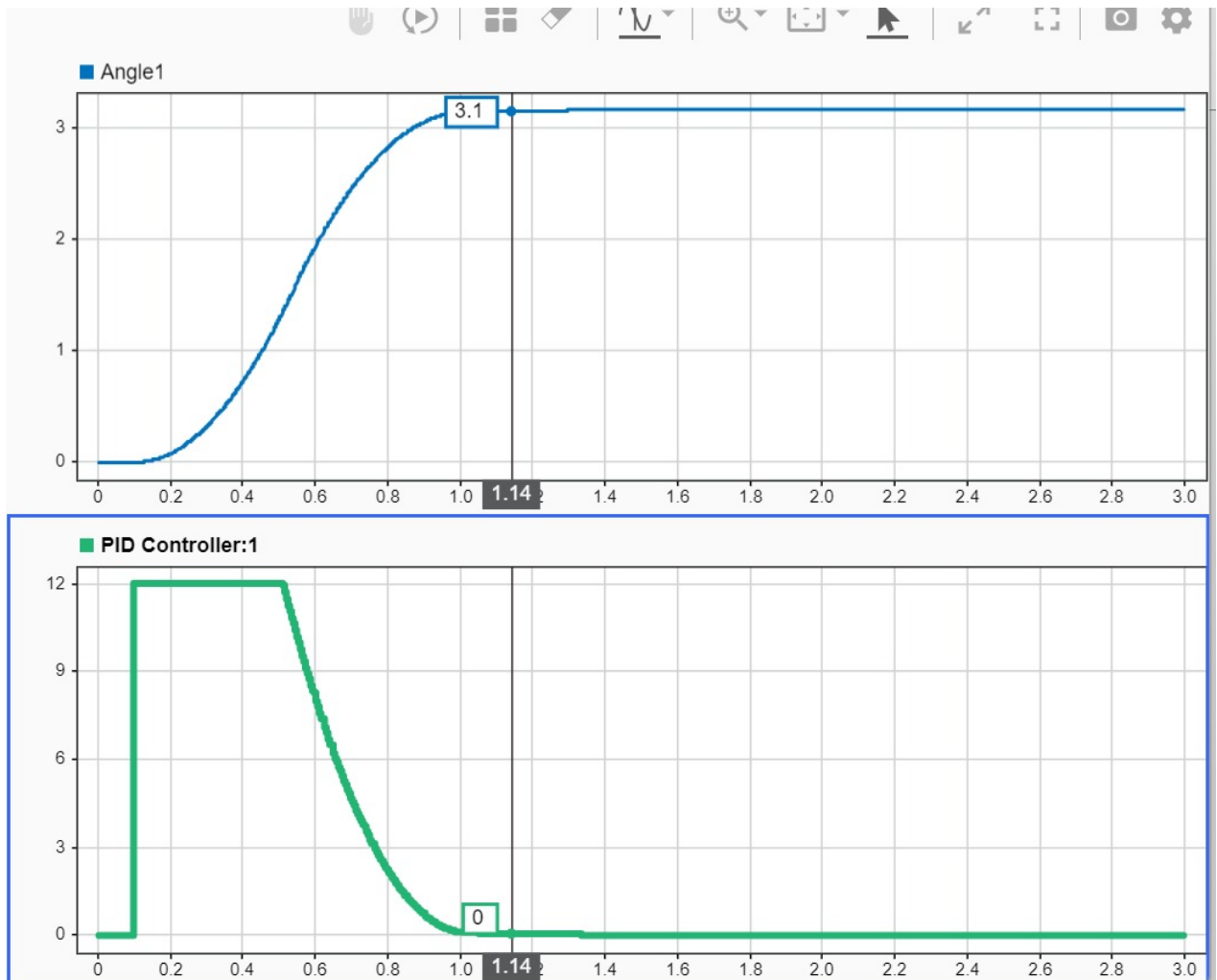
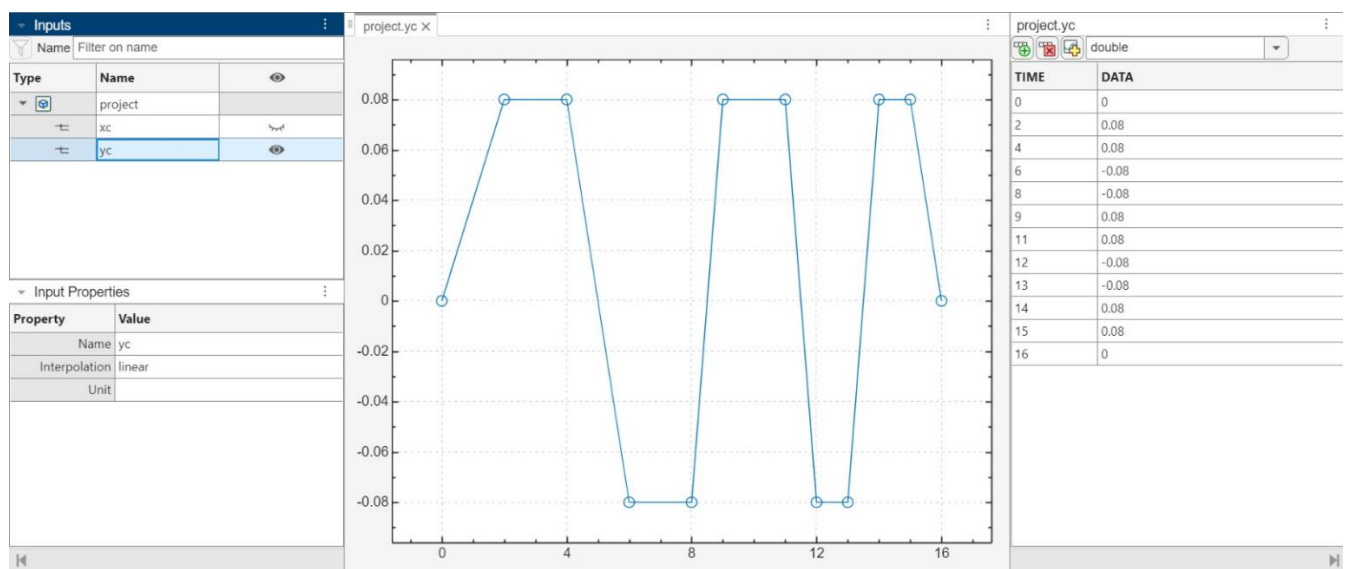
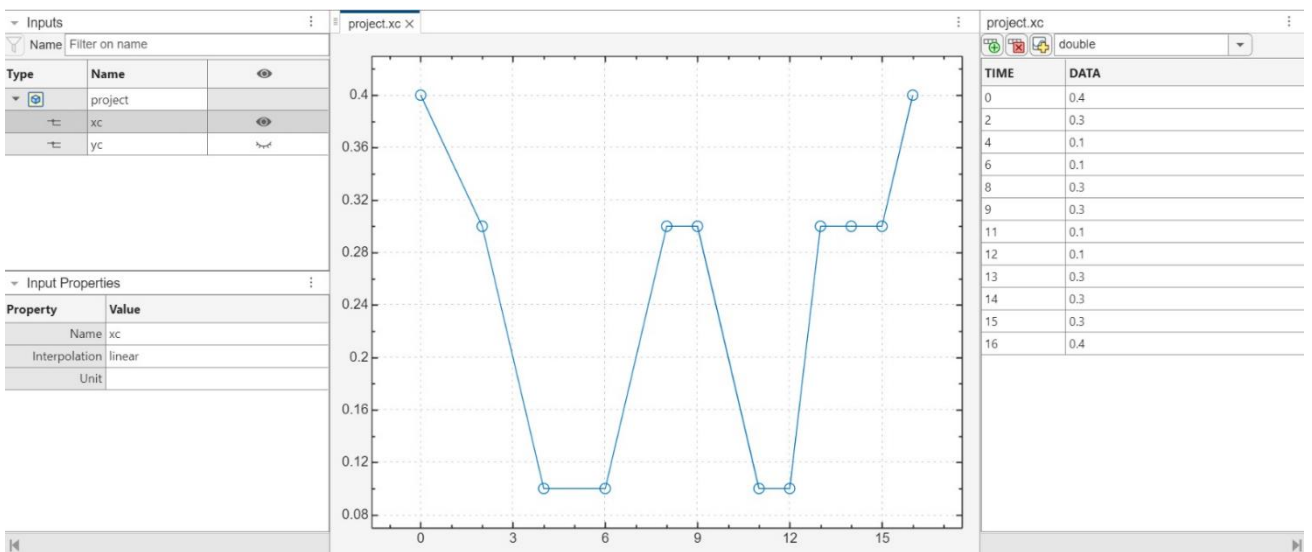


Figure 5

“2-DOF ROBOT ASSEMBLY AND TRAJECTORY POINTS”

‘PHASE 5’

Two identical motor-gear-link joints were assembled to form a 2-DOF planar manipulator. Joint trajectories were generated using inverse kinematics for a square path. The robot traced the square twice: once in 10 seconds and once in 5 seconds.



- **Complete Figure:**

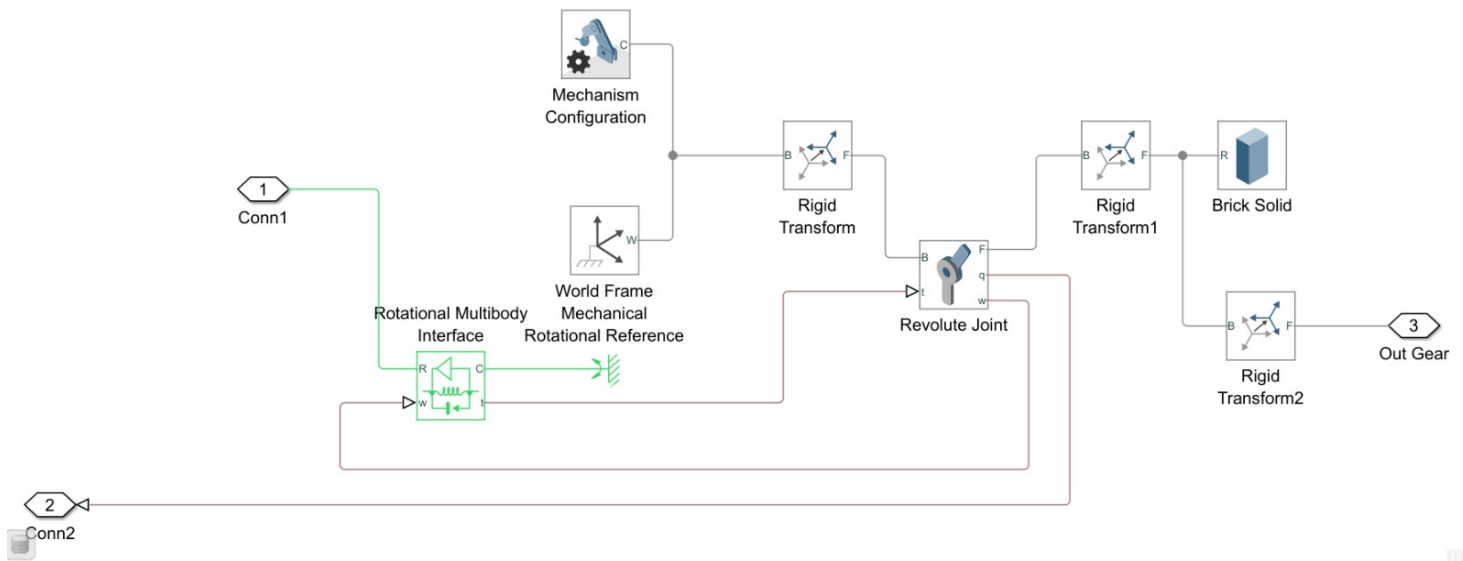
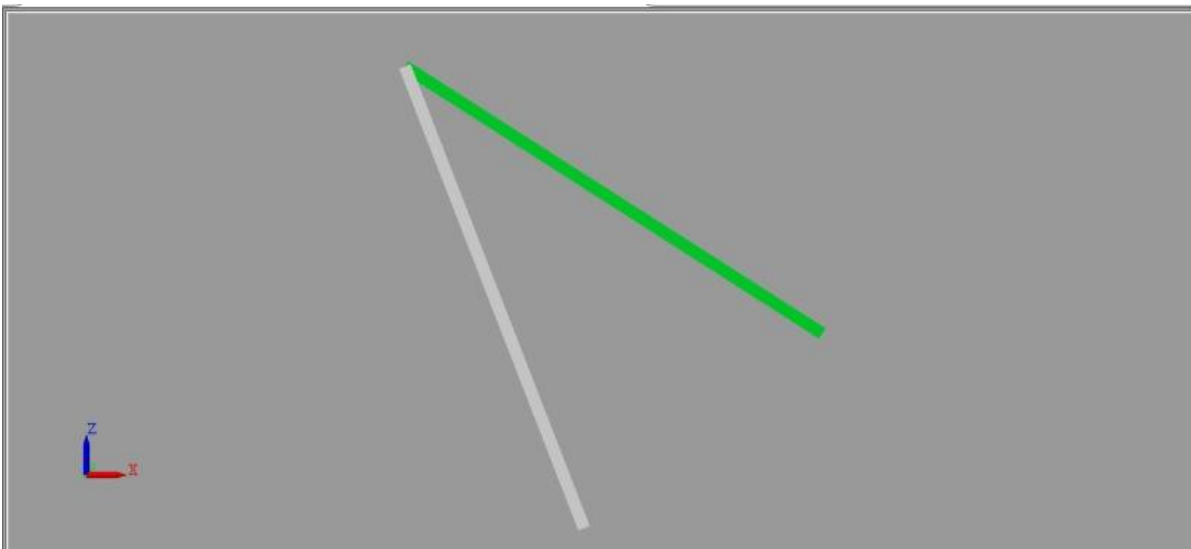
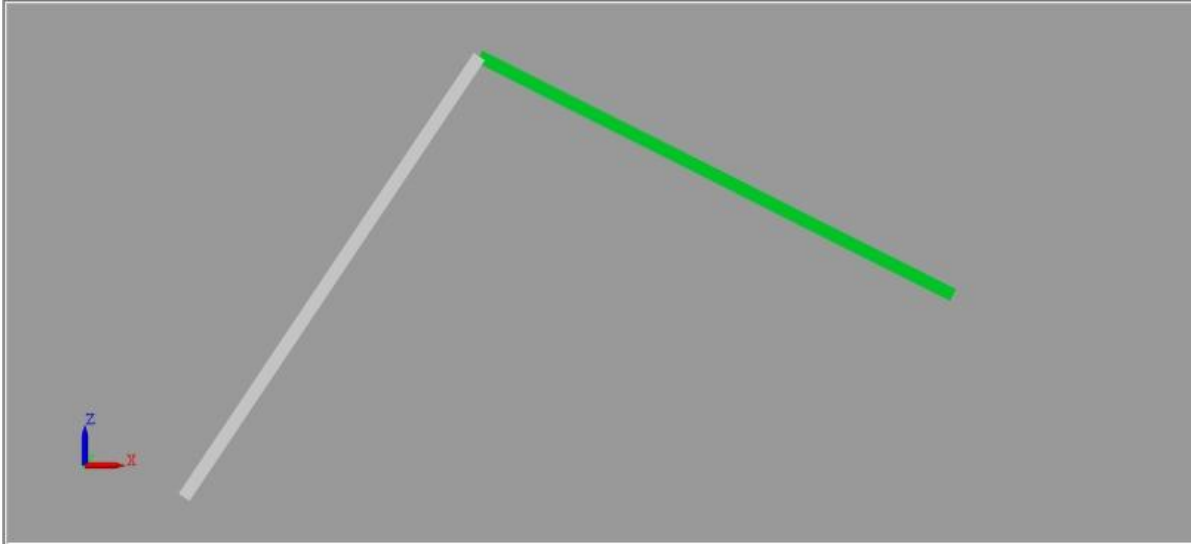


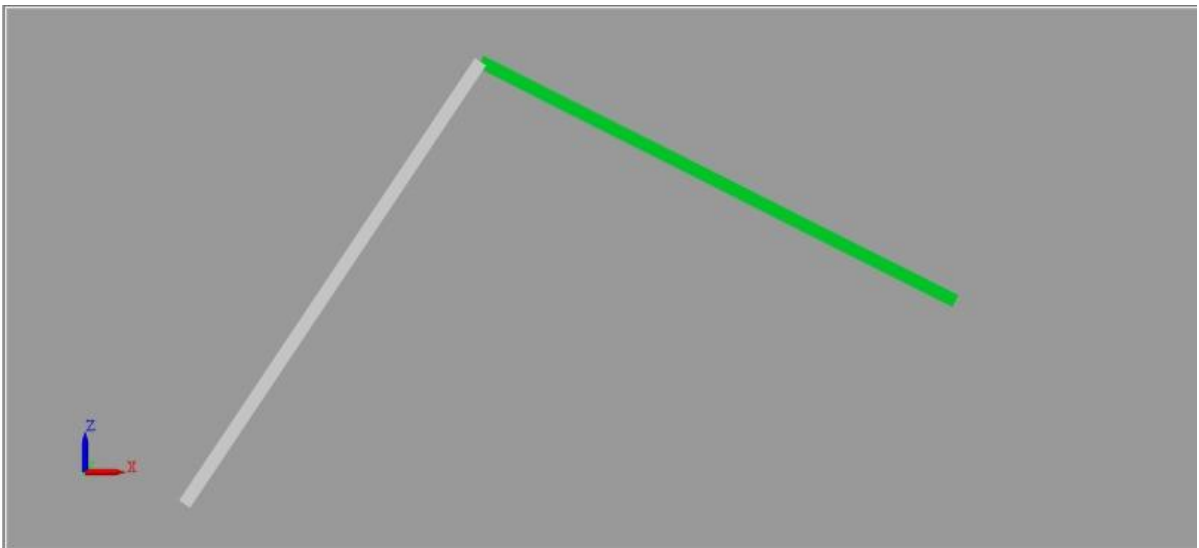
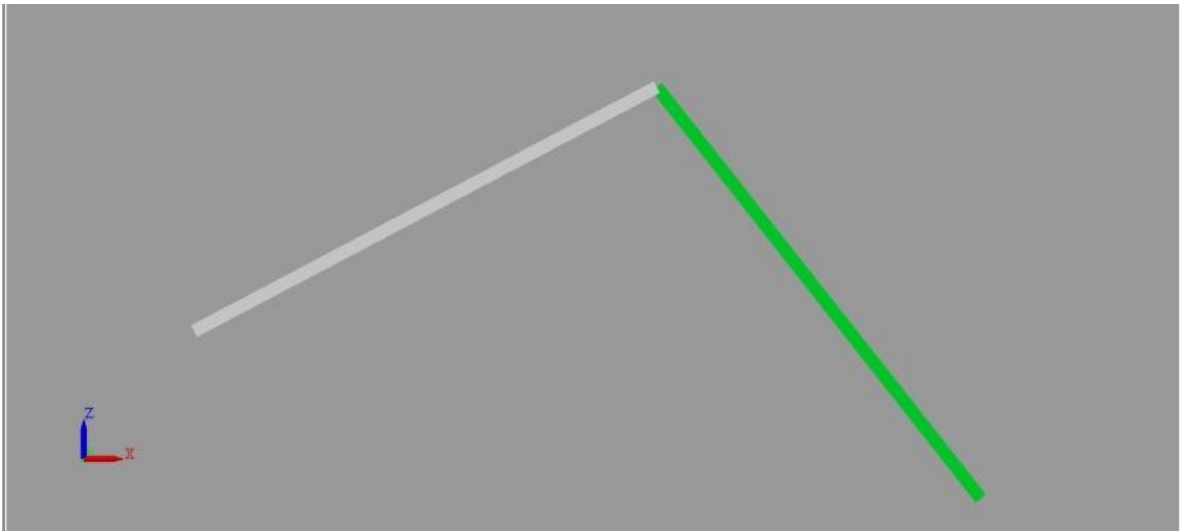
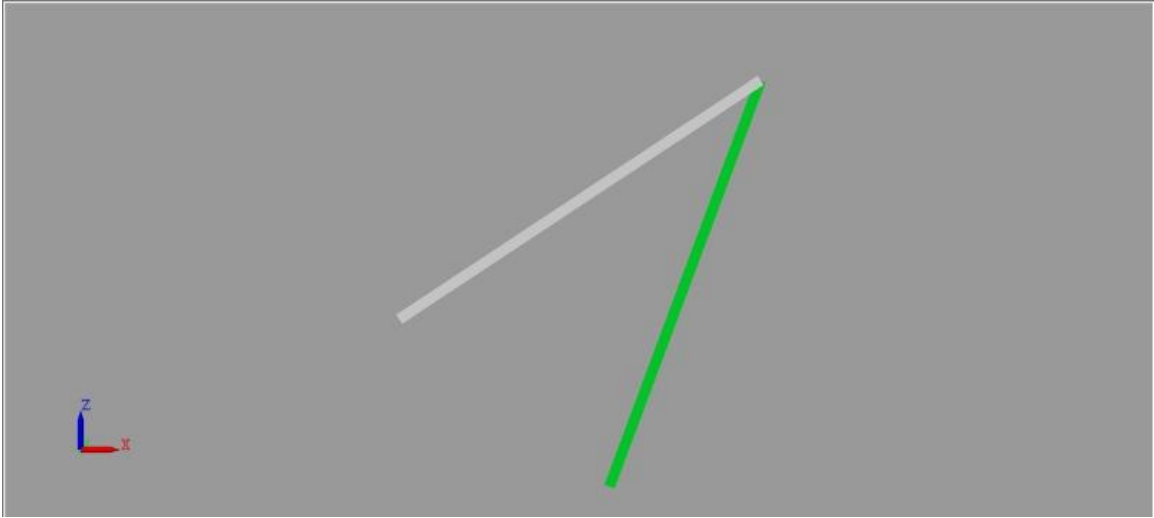
Figure 6

- There was no need of retuning, our same PID values gave us the desired result of the square application on these co-ordinates. These co-ordinates ensures that both the links follows the required trajectories to completely follow the desired application. It completes the first square at 10 seconds approximately and the second one in 5 seconds approximately.

- **Trajectory Tracking:**

Following photos are the trajectory being followed by the link to make a square according to the given co-ordinates.





“POWER ANALYSIS”

‘PHASE 6’

The following quantities were obtained from Calculation:

- Average power consumption
- Peak power requirement
- Copper losses (I^2R)
- Mechanical output power ($\tau\omega$)
- Overall efficiency

6.1 Average Electrical Power Consumption:

The average electrical power consumption was estimated based on the motor current during the execution of the provided trajectory. Since the motor operates under varying load conditions, the average current is significantly lower than the nominal continuous current.

$$P_{\text{avg}} = V \times I_{\text{avg}}$$

Where:

$V = 12\text{V}$ (nominal voltage)

$I_{\text{avg}} = 3.1205\text{ A}$ (This value is avg of no load & nominal current)

$$P_{\text{avg}} = 12 \times 3.1205 = 37.446\text{ W}$$

Thus, the average electrical power consumption of the system is approximately **37 W**.

6.2 Peak Power Requirement:

Peak power occurs during transient conditions such as motor start-up, acceleration, and motion against gravity. These conditions result in a higher current draw for short durations.

$$P_{\text{peak}} = V \times I_{\text{peak}}$$

Where:

$I_{\text{peak}} = 6\text{A}$ (from Datasheet)

$$P_{\text{peak}} = 12 \times 6 = 72 \text{ W}$$

The peak electrical power requirement is therefore approximately **72 W**, occurring only during transient events.

6.3 Copper Loss Analysis (I^2R Losses):

Copper losses in the motor windings were calculated using the armature resistance provided in the datasheet.

$$P_{cu} = I^2 R$$

Where:

$$R = 0.115 \Omega$$

Average Copper Loss:

$$P_{cu,\text{avg}} = (3.1205)^2 \times 0.115 \approx 1.12 \text{ W}$$

Peak Copper Loss:

$$P_{cu,\text{peak}} = (6)^2 \times 0.115 = 4.14 \text{ W}$$

These losses are relatively small due to the low armature resistance of the motor.

6.4 Mechanical Power Output:

The mechanical power delivered at the output joint was calculated using the torque and angular velocity obtained from the trajectory simulation.

$$P_{\text{mech}} = \tau\omega$$

Where:

$$\begin{aligned}\tau &= 3.7 \text{ Nm} \\ \omega &= 6.92 \text{ rad/s}\end{aligned}$$

$$P_{\text{mech}} = 3.7 \times 6.92 = 25.7 \text{ W}$$

Hence, the average mechanical power output of the system is approximately **26 W**.

6.5 Overall System Efficiency:

The overall efficiency of the motor–gear–link system was calculated by comparing the mechanical output power with the electrical input power.

$$\begin{aligned}\eta &= \frac{P_{\text{mech}}}{P_{\text{elec}}} \\ \eta &= \frac{25.7}{37} \approx 0.69\end{aligned}$$

Thus, the overall system efficiency is approximately **69%**.

This value is lower than the motor-only efficiency reported in the datasheet because it includes gear losses, gravitational loading, and dynamic operating conditions.

6.6 Thermal Performance and Temperature Verification:

The thermal performance of the motor was evaluated using the calculated copper losses and the thermal resistance provided in the datasheet.

$$\Delta T = P_{cu,avg} \times R_{\theta}$$

Where:

$$R_{\theta} = 4.7 \text{ K/W}$$

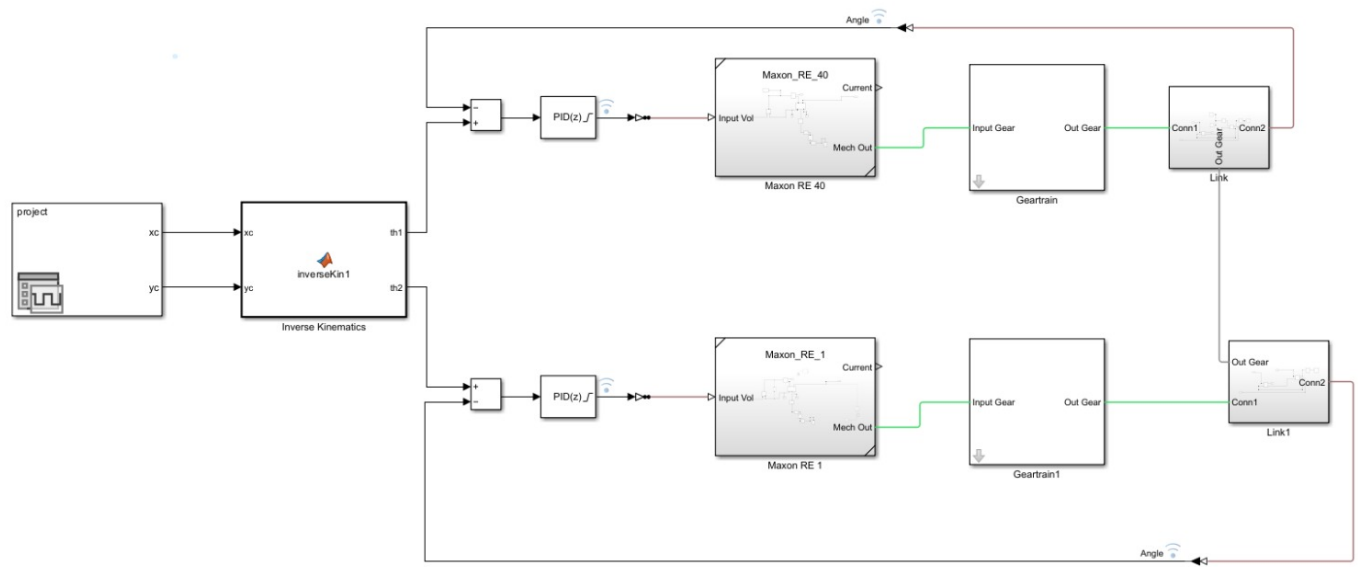
$$\Delta T = 1.12 \times 6.6 \approx 5.2624^{\circ}\text{C}$$

With an ambient temperature of 25°C , the estimated winding temperature is:

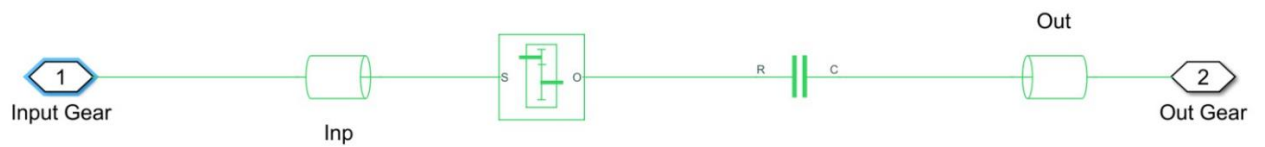
$$T \approx 30.2624^{\circ}\text{C}$$

- This temperature is well below the maximum allowable winding temperature of **155°C** , confirming safe continuous operation without the need for an external heat sink.

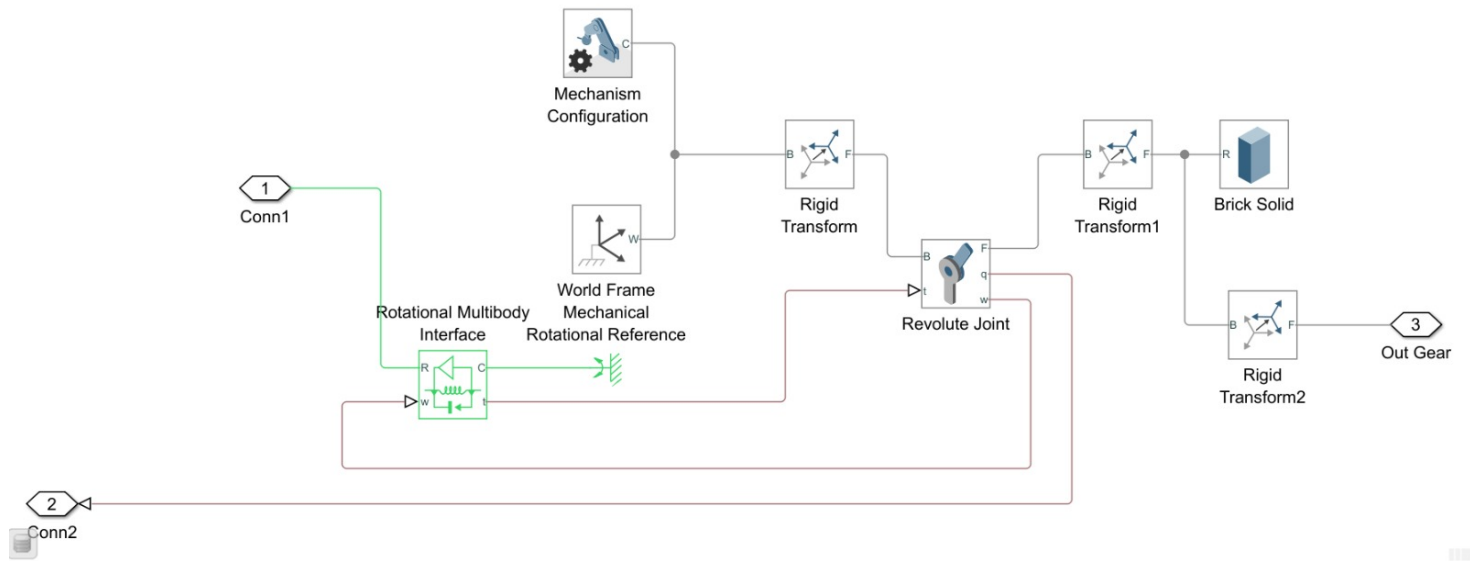
“COMPLETE MODEL”



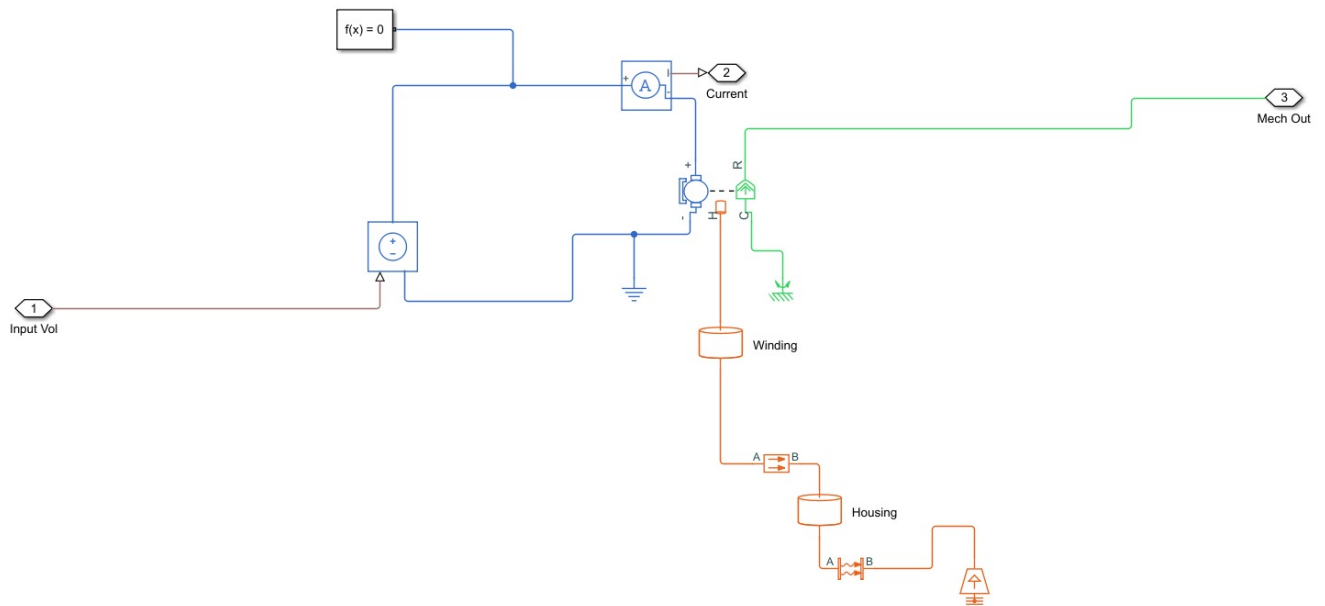
COMPLETE MODEL



GEAR-TRAIN



LINK



MAXON RE 40 (DC MOTOR)

“CONCLUSIONS & FUTURE WORK”

A. Conclusions:

The proposed project successfully realized the design, modeling, and simulation process for a 2 DOF planar robotic manipulator. By incorporating the high-performance Maxon RE 40 (148866) motor in connection with the LHT-20-100-U-III strain wave gear, a robust joint for the manipulator was developed.

➤ Key achievements include:

- **System Integration:** A unified SimScape model was developed, effectively bridging the electrical, mechanical, and thermal domains.
- **Model Validation:** The motor model's no-load speed and stall torque were validated against the datasheet, while the thermal subsystem correctly predicted winding temperature rises within the **+155°C** limit.
- **Precision Control:** Discrete PID controllers were tuned to meet rigorous performance criteria, achieving a settling time of **< 1 seconds** and steady-state error of **< 1 degree** for any step input.
- **Trajectory Fidelity:** The assembled 2-DOF system successfully traced complex square trajectories in 10-second and 5-second intervals, maintaining tracking accuracy despite gravitational disturbances.

B. Limitations:

- **Simplified Friction Model:** Even though static and Coulomb friction were implemented, the model does not include the position-dependent or temperature-dependent variations in friction that might be present in strain wave gears.
- **Rigid Link Assumption:** Ideal rigid bodies were assumed for the robot links. This model does not account for what would be, in a physical

system, the most expected influence of the high torque from the Maxon RE 40, such as structural vibrations or elastic deflections.

- **Ideal Power Electronics:** The simulation is ideal because it treats the voltage source and motor driver as perfect, without any kind of switching loss or EMI that would be present in physical controllers, this also effects output Torque conflation in calculation and real time value, also in the whole evaluation.
- **Lack of Experimental Validation:** Results are purely computational. In order to verify the calculated thermal mass and efficiency of the heat sink design, a physical prototype is needed.

C. Future Work:

These amendments were offered by the LLM model too and also reviewed to showcase our challenges.

- **Advanced Nonlinear Modeling:** The future models should include backlash and hysteresis for a more accurate representation of the harmonic drive because even precision gearboxes can have nonlinearities in these aspects.
- **Effective Control Strategies:** Robust control strategies such as the use of the Adaptive Control method and Sliding Mode Control would be an improvement over the current strategy of using a PID controller, which would be less effective than the proposed strategy at adapting to the varied weight and external forces.

- **Thermal Optimization:** Further research in active cooling technologies may potentially enable the motor to run beyond its current continuous time limit by pursuing an aggressive winding-to-housing heat transfer approach.
- **Hardware-in-the-Loop (HIL);** The transition of the proven Simulink model to the real-world hardware testing platform would help the final jump towards the physical 2-DOF robotic system implementation.

“APPENDICES”

- **Code Listings & Datasheets:**

- Maxon Motor, RE 40 DC Motor Datasheet was used.
- Only code of inverse kinematics was used which is given below:

```
function [th1,th2] = inverseKin1(xc,yc, l1, l2)

D = (xc^2 + yc.^2 - l1^2 - l2^2)/(2*l1*l2);
th2 = atan2(-sqrt(1-D^2),D);
th1 = atan2(yc,xc) -
atan2(l2*sin(th2), (l1+l2*cos(th2)));
```

References

1. Maxon Motor, RE 40 DC Motor Datasheet.
2. Harmonic Drive, LHT-20-100-U-III Datasheet.
3. ChatGPT.
4. Gemini.